



SCEGGS Darlinghurst

2003
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 84

- Attempt Questions 1–7
- All questions are of equal value

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Centre Number

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Student Number

Total marks – 84

Attempt Questions 1–7

All questions are of equal value

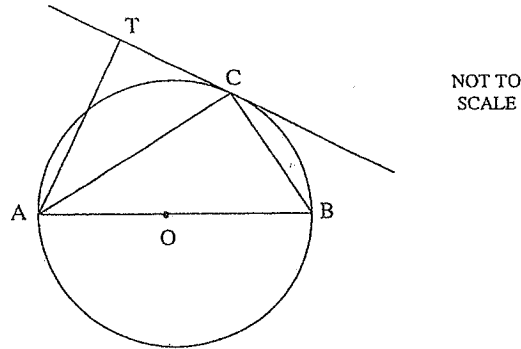
Answer each question on a NEW page.

	Marks
Question 1 (12 marks)	
(a) Find the co-ordinates of the point P which divides the interval joining A(–3, 2) and B(5, 6) externally in the ratio 1:3.	2
(b) Differentiate $x \tan^{-1}(x^2)$.	2
(c) Find n if $\binom{n}{2} = 91$.	2
(d) Evaluate $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$.	2
(e) Given $x+3$ is a factor of $P(x) = x^3 - Ax^2 + 2x - 1$. Find the value of A.	2
(f) Explain why $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \frac{2}{3}$	2

Marks

Question 2 (12 marks) Start a NEW page.

(a)



3

AOB is the diameter of a circle centre O and C is the point of contact of the tangent TC such that AC bisects $\angle TAB$. Prove that AT is perpendicular to TC.

(b) How many arrangements of the letters of the word DEFINITION are there if the letters N are not together?

2

(c) Consider the function

$$y = \frac{1}{2} \cos^{-1}(2x - 1)$$

(i) Find its domain.

1

(ii) Sketch the function.

1

(iii) Evaluate y if $x = \frac{1}{4}$.

2

(d) When a biased coin is tossed it shows heads in 2 out of every 3 tosses. The coin is tossed 15 times. Find:

(i) the probability of 12 heads.

1

(ii) the probability of at least 2 heads.

2

You may leave your answers in index form.

Question 3 (12 marks) Start a NEW page.

Marks

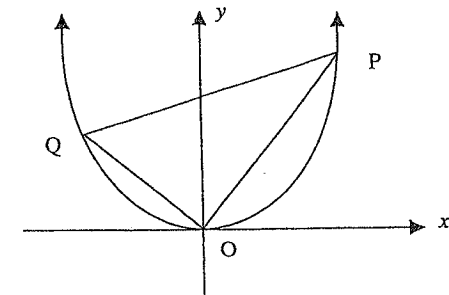
(a) Find the co-efficient of x in the expansion of $\left(x^2 - \frac{3}{x}\right)^8$.

3

(b) Evaluate $\int_0^3 \frac{x}{\sqrt{1+x}} dx$ using the substitution $x = u^2 - 1$.

3

(c)



P(2ap, ap²) and Q(2aq, aq²) are points on the parabola $x^2 = 4ay$. It is given that $\angle POQ = 90^\circ$. O is the origin.

(i) Prove that $pq = -4$

2

(ii) Find the co-ordinates of M, the midpoint of PQ.

1

(iii) Prove that the Cartesian equation of the locus of M is

3

$$2ay = x^2 + 8a^2.$$

Marks

Question 4 (12 marks) Start a NEW page.

(a) If the equation $3x^3 - 4x^2 + 2x + 1 = 0$ has roots α , β and γ find:

(i) $2\alpha + 2\beta + 2\gamma$.

1

(ii) the equation whose roots are 2α , 2β and 2γ .

2

(b) A metal sphere is heated such that the surface area is increasing at $4\pi \text{ mm}^2$ per minute.

(i) Find the rate of increase of the radius when the radius is 20mm.

3

(ii) Find the rate of increase of the volume at this time.

2

(c) Use Mathematical Induction to prove that

4

$$2 \times 5 + 4 \times 8 + \dots + 2n(3n+2) = n(n+1)(2n+3)$$

for all positive integers n .

Marks

Question 5 (12 marks) Start a NEW page.

(a) (i) Prove that $\frac{1}{1+\cos x} = \frac{1}{2} \sec^2 \frac{x}{2}$

2

(ii) Hence evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{1+\cos x}$

2

(b) A committee of 5 is to be formed from a group of 9 men and 7 women.

(i) How many different committees are possible?

1

(ii) How many would include only 3 women?

1

(iii) How many are possible if Mr and Mrs Brown cannot both serve?

2

(c) Use the substitution $u = \ln x$ to evaluate

4

$$\int_1^e \frac{dx}{x(1+2\ln x)^2}$$

Marks

Question 6 (12 marks) Start a NEW page.

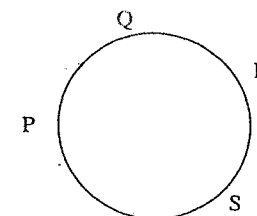
- (a) (i) Sketch $y = f(x)$ if $f(x) = e^{x+2}$ 1
- (ii) Find the inverse function $f^{-1}(x)$. 2
- (iii) Sketch $y = f^{-1}(x)$. You may choose to do this on your sketch in part (i). 1
- (b) A particle moves in a straight line so that when it is x metres from the origin O its velocity v m/s is given by
- $$v^2 = 32 + 8x - 4x^2$$
- (i) Prove that the particle is moving in Simple Harmonic Motion. 2
- (ii) Find the centre of the motion. 1
- (iii) Find the period and amplitude of the motion. 2
- (iv) Given that the particle is initially at $x = 4$, which of the 2 equations could describe its motion: 1
- $$x = 1 + 3 \sin 2t$$
- or
- $$x = 1 + 3 \cos 2t$$
- (v) When does the particle pass through the origin for the first time? 2

Marks

Question 7 (12 marks) Start a NEW page.

- (a) Evaluate exactly $\sin\left(2\cos^{-1}\frac{2}{3}\right)$ 2
- (b) (i) Express $\sqrt{3}\sin x - \cos x$ in the form $A\sin(x - \alpha)$ if $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ 2
- (ii) Hence sketch the curve 3
- $$y = \sqrt{3}\sin x - \cos x \text{ for } 0 \leq x \leq 2\pi$$
- showing all important features.
- (iii) Use your sketch to determine the value(s) of k for which 1
- $$\sqrt{3}\sin x - \cos x = k$$
- has 3 distinct solutions for $0 \leq x \leq 2\pi$

(c)



NOT TO
SCALE

P, Q, R and S are points on the circumference of a circle.

Prove that $\frac{PR}{QS} = \frac{\sin \angle PQR}{\sin \angle QPS}$

END OF PAPER

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1) a) A(-3,2) B(5,6) ratio 1:3

$$x = \frac{-3 \times 3 + 5 \times 1}{-1 + 3}$$

$$= \frac{-14}{2}$$

$$y = \frac{2 \times 3 + 6 \times 1}{2}$$

$$= 0$$

$$\therefore P \text{ is } (-7, 0)$$

b) $\frac{d}{dx} [x \tan^{-1}(x^2)]$

$$= \tan^{-1}(x^2) + \frac{x \times 2x}{1+x^2}$$

$$= \tan^{-1}(x^2) + \frac{2x^2}{1+x^2}$$

c) $\frac{n!}{(n-2)!2!} = 91$

$$(n-2)!2!$$

$$\therefore n(n-1) = 182$$

$$n^2 - n - 182 = 0$$

$$(n-14)(n+13) = 0$$

$$n = 14$$

d) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_0^1$

$$= \sin^{-1} \frac{1}{2} - \sin^{-1} 0$$

$$= \frac{\pi}{6}$$

e) $P(-3) = -27 - 9A - 6 - 1 = 0$

$$\therefore 9A = -34$$

$$A = -3\frac{7}{9}$$

f) $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \times \frac{2}{3}$

since $\lim_{x \rightarrow 0} \frac{\sin 2x}{2x} = 1$

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$$\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = 1 \times \frac{2}{3} = \frac{2}{3}$$

(2) a)

$\angle TCA = \angle CBA$ (angle between tangent and chord equals angle in alternate segment)

$\angle TAC = \angle CAB$ (AC bisects $\angle TAB$, given)

$\angle ACB = 90^\circ$ (angle in semi circle is 90°)

$\triangle TAC \cong \triangle ACB$ (equiangular)

$\therefore \angle ATC = \angle ACB$ (corresp $\angle$'s in sim $\triangle$'s)

$\therefore AT \perp TC$

b) no. with H's together = $\frac{9!}{3!}$

total no. = $\frac{10!}{3!2!}$

$\therefore$ no. with H's apart = $\frac{10!}{2!3!} - \frac{9!}{3!}$

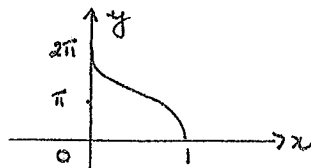
$$= 241920$$

c) (i) $-1 \leq 2x-1 \leq 1$

$$0 \leq 2x \leq 2$$

Domain: $0 \leq x \leq 1$

(ii)



(iii) $y = \frac{1}{2} \cos^{-1}(\frac{1}{2} - 1)$

$$= \frac{1}{2} \cos^{-1}(-\frac{1}{2})$$

$$= \frac{1}{2} (\pi - \frac{\pi}{3})$$

$$= \frac{\pi}{3}$$

d) $P(H) = P = \frac{2}{3}$

$$P(T) = q = \frac{1}{3}$$

(i) $P(12H) = \binom{15}{12} \left(\frac{2}{3}\right)^{12} \left(\frac{1}{3}\right)^3$

$$P(\text{at least 2H}) = 1 - P(0H) - P(1H)$$

$$= 1 - \binom{15}{0} \left(\frac{1}{3}\right)^{15} - \binom{15}{1} \left(\frac{1}{3}\right)^{14} \left(\frac{2}{3}\right)^1$$

(3) a) $T_{k+1} = \binom{8}{k} (x^2)^{8-k} (-3x^{-1})^k$

$$= \binom{8}{k} x^{16-2k} (-3)^k x^{-k}$$

$$= \binom{8}{k} (-3)^k x^{16-3k}$$

$$\therefore 16-3k=1$$

$$3k=15$$

$$k=5$$

coefficient is $\binom{8}{5} (-3)^5$

$$= -13608$$

b) $x = u^2 - 1$

$$dx = 2u du$$

if $x=3$, $u=2$

$x=0$, $u=1$

$$\int_0^3 \frac{x dx}{\sqrt{x+1}} = \int_1^2 \frac{u^2-1}{u} \times 2u du$$

$$= 2 \int_1^2 (u - \frac{1}{u}) du$$

$$= 2 \left[\frac{u^2}{2} - \ln u \right]_1^2$$

$$= 2 \left(\frac{8}{2} - 2 - \frac{1}{2} + 1 \right)$$

$$\int_0^3 \frac{x dx}{\sqrt{x+1}} = \frac{8}{3}$$

c) (i) gradient $PO = \frac{ap^v}{2ap} = \frac{p}{2}$

gradient $QO = \frac{q}{2}$

since $PO \perp OQ$, $\frac{p}{2} \times \frac{q}{2} = -1$

$$pq = -4$$

(ii) M is $(ap+aq, \frac{ap^v+aq^v}{2})$

(iii) $x^v = a^v p^v + a^v q^v + 2a^v p q$
 $= a^v p^v + a^v q^v - 8a^v$

$$\therefore x^v + 8a^v = a^v p^v + a^v q^v$$

$$2ay = 2a \cdot \frac{ap^v+aq^v}{2}$$

$$= a^v p^v + a^v q^v$$

$$\therefore 2ay = x^v + 8a^v \text{ is locus of M}$$

(4) a) (i) $2 + \beta + \gamma = \frac{4}{3}$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{2}{3}, \quad \alpha\beta\gamma = -\frac{1}{3}$$

$$\therefore 2(\alpha + \beta + \gamma) = \frac{8}{3}$$

(ii) equation is:-

$$x^3 - (2\alpha + 2\beta + 2\gamma)x^2 + (4\alpha\beta + 4\alpha\gamma + 4\beta\gamma)x - \alpha\beta\gamma = 0$$

$$\text{or } 3x^3 - 8x^2 + 8x + 8 = 0$$

c) Consider $n=1$,

$$L.H.S. = 2 \times 5 = 10$$

$$R.H.S. = 1(2)(5) = 10$$

$$\therefore \text{true for } n=1$$

assume true for $n = k$

$$\therefore 2 \times 5 + 4 \times 8 + \dots + 2k(3k+2) = k(k+1)(2k+3)$$

consider $n = k+1$

$$\begin{aligned} L.H.S. &= 2 \times 5 + \dots + 2k(3k+2) + 2(k+1)(3k+5) \\ &= k(k+1)(2k+3) + 2(k+1)(3k+5) \\ &= (k+1)[2k^2 + 3k + 6k + 10] \\ &= (k+1)(2k^2 + 9k + 10) \\ &= (k+1)(k+2)(2k+5) \\ &= R.H.S. \text{ if } n = k+1. \end{aligned}$$

if true for $n = k$ it is also true for $n = k+1$. It is true for $n=1$ and thus for $n=2, 3, \dots$ i.e. true for all n positive integers.

⑤ a) (i) $1 + \cos 2x = 2 \cos^2 x$
 $\therefore 1 + \cos x = 2 \cos^2 \frac{x}{2}$

$$\therefore \frac{1}{1 + \cos x} = \frac{1}{2} \sec^2 \frac{x}{2}$$

(ii) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{dx}{1 + \cos x} = \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$
 $= \left[\tan \frac{x}{2} \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$
 $= \tan \frac{\pi}{4} - \tan \frac{\pi}{6}$
 $= 1 - \frac{1}{\sqrt{3}}$

b) (i) no. possible = $\binom{16}{5} = 4368$

(ii) no. (3 women) = $\binom{7}{3} \times \binom{9}{2} = 1260$

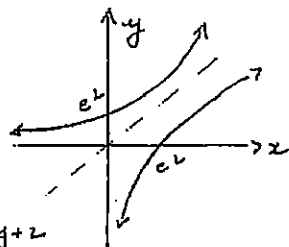
(iii) no. with both = $\binom{14}{3}$

$\therefore$ no required = $4368 - 1260 = 3108$

c) $u = \ln x$ $x=4, u=1$
 $du = \frac{1}{x} dx$ $x=1, u=0$

$$\begin{aligned} \int_1^e \frac{dx}{x(1+2\ln x)^2} &= \int_0^1 \frac{du}{(1+2u)^2} \\ &= \left[-\frac{1}{2}(1+2u)^{-1} \right]_0^1 \\ &= \left[-\frac{1}{2(1+2u)} \right]_0^1 \\ &= -\frac{1}{6} + \frac{1}{2} \\ &= \frac{1}{3} \end{aligned}$$

⑥ a) (i)



(ii) $x = e^{y+2}$
 $y+2 = \ln x$
 $y = \ln x - 2$

b) $v^2 = 32 + 8x - 4x^2$

(i) $\frac{1}{2} v^2 = 16 + 4x - 2x^2$
 $\frac{d}{dt} \left(\frac{1}{2} v^2 \right) = \ddot{x} = 4 - 4x$
 $= -4(x-1)$

which is of the form of S.H.M.
 $\ddot{x} = -\omega^2(x-x_1)$

(ii) centre is $x=1$

(iii) period $T = \pi$ seconds

if $v=0$, $4(x^2 - 2x - 8) = 0$
 $(x-4)(x+2) = 0$

$x = 4, -2$ when $v=0$

$\therefore$ amplitude = 3m

(iv) testing $x = 1 + 3 \sin 2t$
 when $t=0$, $x=1$

testing $x = 1 + 3 \cos 2t$

when $t=0$, $x=4$

$\therefore x = 1 + 3 \cos 2t$ could describe the motion.

(v) $0 = 1 + 3 \cos 2t$

$3 \cos 2t = -1$

$\cos 2t = -\frac{1}{3}$

Acute angle = 1.231 (4 s.f.)

2nd 3rd quadrants

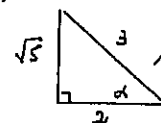
$\therefore 2t = \pi - 1.231$

$= 1.9106 \dots$

$\therefore$ first time 0.955 s (approx)

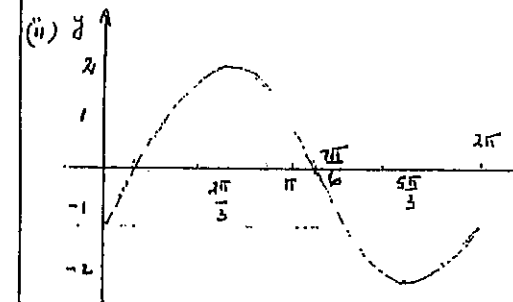
⑦ a) let $\cos^{-1} \frac{2}{3} = \alpha$

$\therefore \sin 2\alpha = 2 \sin \alpha \cos \alpha$
 $= 2 \times \frac{\sqrt{5}}{3} \times \frac{2}{3}$
 $= \frac{4\sqrt{5}}{9}$



b) (i) $\sqrt{3} \sin x - \cos x$
 $= A \sin x \cos \phi - A \cos x \sin \phi$
 $A = 2, \therefore \cos \phi = \frac{\sqrt{3}}{2}$
 $\phi = \frac{\pi}{6}$

$\therefore \sqrt{3} \sin x - \cos x = 2 \sin \left(x - \frac{\pi}{6} \right)$



(iii) 3 solutions if $k = -1$

c) In ΔPQR ,
 $\frac{PR}{\sin \angle QPR} = \frac{QR}{\sin \angle QPR}$ (Sine Rule)

$\angle QPR = \angle QSR$ (angles in same segment are equal)

$\therefore \frac{PR}{\sin \angle QPR} = \frac{QR}{\sin \angle QSR}$

In ΔQRS ,
 $\frac{QR}{\sin \angle QSR} = \frac{QS}{\sin \angle QRS}$ (Sine Rule)

$\therefore \frac{PR}{\sin \angle QPR} = \frac{QS}{\sin \angle QRS}$

$\therefore \frac{PR}{QS} = \frac{\sin \angle QPR}{\sin \angle QRS}$

but $\angle QRS = 180^\circ - \angle QPS$

(opp angles of a cyclic quadrilateral are supplementary)

$\therefore \sin \angle QRS = \sin \angle QPS$

$\therefore \frac{PR}{QS} = \frac{\sin \angle QPR}{\sin \angle QPS}$